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RESEARCH

Multi-scale quadratic convolutional neural network for bearing fault diagnosis based on multi-sensor data fusion

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Abstract Bearing fault diagnosis is crucial for the safe and stable operation of mechanical equipment. However, bearing signals are highly susceptible to noise interference, which complicates feature extraction. Existing multi-source data diagnostic methods still face challenges in effectively integrating signals and suppressing noise. To address these challenges, this paper proposes a multi-sensor data fusion and multi-scale quadratic convolutional neural network for intelligent bearing fault diagnosis. First, the method inputs vibration signals collected by multiple sensors into a time domain filter consisting of a quadratic convolutional network and a frequency domain filter based on a fully connected neural network for processing. The filtered signal is then passed into a multi-scale quadratic convolutional neural network, which utilizes quadratic neurons with strong feature extraction capabilities for bearing vibration signals. The extracted multi-scale features are further refined through a cross attention mechanism to capture more useful information, which is then classified. Experimental results conducted on the bearing datasets from Case Western Reserve University and Politecnico di Torino demonstrate that the proposed method outperforms other comparative models in noisy environments. At a signal-to-noise ratio of −10, the method achieves accuracies of 97.20% and 98.81%, respectively, verifying its excellent performance under complex noise interference conditions.

Keywords Quadratic convolution · Deep learning · Filtering · Multi-scale feature extraction · Fault diagnosis

1 Introduction

With the continuous advancement of industrial automation, rotating machinery such as aircraft engines, pumps, and wind turbines are playing an increasingly critical role in various industrial applications [1]. Bearings serve multiple functions in mechanical equipment, including supporting, reducing friction, and transmitting torque, making them essential for the proper operation and performance of machinery [2,3]. The quality and performance of bearings directly impact the reliability and service life of industrial equipment. However, bearing failures, such as corrosion, wear, and fractures, are inevitable during the operation of industrial systems [4,5]. These failures can lead to abnormal vibrations, noise, and other disturbances, which may reduce production efficiency and even jeopardize personnel safety. Therefore, regular inspection and fault diagnosis of bearings are essential for predicting potential issues and ensuring the stable operation of equipment. This is crucial for maintaining the stability of industrial production, improving safety, and promoting sustainable development [6–9].

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The various sensing signals generated during the operation of rolling bearings are invaluable for mechanical fault diagnosis. These signals include vibration signals, acoustic emission signals, and current signals [10,11]. Among them, using acceleration sensors to collect vibration signals generated by bearing rotation is the most commonly employed method for signal acquisition. Traditional machine learning-based mechanical fault diagnosis methods typically involve three steps: data acquisition, feature extraction, and fault classification. Of these, feature extraction is the key to accurate fault diagnosis [12–14]. Traditional feature extraction methods include Fourier Transform (FT), Wavelet Transform (WT), Variational Mode Decomposition (VMD), and Feature Mode Decomposition (FMD) [15,16]. While these methods have contributed to fault diagnosis, they are heavily reliant on expert knowledge and, due to the shallow structure of machine learning models, have limitations in handling nonlinear problems, improving generalization, and enhancing robustness [17]. As a result, they fail to fully meet the current demands of intelligent fault diagnosis.

Compared to traditional machine learning methods, deep learning has become widely adopted in the field of bearing fault diagnosis due to its powerful deep nonlinear structure and excellent feature extraction capabilities [18–20]. Deep learning methods help eliminate the tedious process of manual feature extraction, enabling end-to-end intelligent diagnosis and improving both diagnostic efficiency and accuracy. Common deep learning methods include Variational Autoencoders (VAE), Long Short-Term Memory networks (LSTM), Deep Belief Networks (DBN), and Convolutional Neural Networks (CNN) [21–24]. Among these, CNNs have been widely applied in fault diagnosis due to their flexible structure and robust multi-level nonlinear representation abilities, achieving significant results.Zhang et al. [25] proposed a deep convolutional neural network (WDCNN) with a wide first-layer kernel, which delivered promising results in bearing fault diagnosis. Yu et al. [26] developed a Broad Convolutional Neural Network (BCNN) with incremental learning capability, which enhances the model's diagnostic performance by adding new fault samples. Qin et al. [27] used continuous wavelet transform time-frequency maps combined with lightweight convolutional neural networks for fast, online fault diagnosis of bearings. He et al. [28] proposed an intelligent fault diagnosis method, ECCR-MFFN, which demonstrates excellent diagnostic performance under imbalanced data conditions. Xu et al. [29] developed an attention-based multi-receptive field convolutional neural network and designed an adaptive label adjustment loss, enabling accurate fault identification even in scenarios with imbalanced data distribution. Li et al. [30] addressed the issue of imbalanced classification in bearing fault data by proposing a central jump boosting method for semi-supervised and imbalanced data-driven intelligent fault recognition.

However, due to the susceptibility of bearings to noise interference during operation, using CNNs alone for feature extraction from noisy vibration signals may lead to a decrease in diagnostic accuracy. Therefore, it is essential to enhance the network's feature extraction capabilities. To address this issue, Jiang et al. [31] introduced a multi-head attention mechanism that improved the aggregation of bearing fault features, significantly enhancing the feature extraction performance of the network. Wei et al. [32] combined CNN with a self-attention mechanism and proposed the WSAFormer-DFFN model for fault diagnosis. Experimental results show that this model achieves high recognition accuracy and strong performance stability in noisy environments. Additionally, Li et al. [33] designed a multi-branch denoising convolutional network that enhances bearing fault diagnosis accuracy by filtering out noise components and extracting multi-level feature representations. Huang et al. [34] combined multi-scale learning with attention mechanisms to improve the network's feature extraction ability, demonstrating its effectiveness in bearing fault diagnosis through experimental analysis in noisy environments. Huo et al. [35] proposed a multi-scale parallel feature extraction method to fully exploit fault features, enabling unsupervised bearing fault diagnosis. Niazi et al. [36] proposed a TT-ConvLSTM neural network technique that improves the model's efficiency and generalization by employing parallel computing and sliding window techniques. These methods effectively enhance the fault diagnosis capabilities of deep learning models under noise interference.

The methods discussed in the above literature are all based on single-sensor data. However, key components such as rolling bearings often operate in harsh environments, where a single sensor provides limited information and is susceptible to issues such as noise pollution and missing data. Additionally, vibration sig-

nals captured from different sensor positions are typically non-stationary and interdependent. To overcome the limitations of single-sensor signals in terms of performance and operating conditions, intelligent fault diagnosis must incorporate multi-sensor information fusion. This approach improves the accuracy and reliability of fault diagnosis by extracting more comprehensive and richer information.Guan et al. [37] proposed a weighted strategy based on kurtosis criteria to fuse sensor data from three directions into a single dataset. Zhu et al. [38] concatenated four one-dimensional sensor signals into a new signal and used a deep Q-network to learn the combination of the new signal. Experimental results show that this method performs exceptionally well under small sample conditions. Xu et al. [39] developed a graph-guided collaborative convolutional neural network composed of two multi-scale feature extraction networks (MFEN) and a graph reasoning fusion network (GRFN), which can effectively identify faults. However, these fusion methods have limitations in dealing with strong noise environments. When the fault signal is masked by strong noise, the feature learning performance of these methods significantly decreases. Therefore, fault diagnosis still faces challenges in noisy conditions.To address these challenges and improve recognition accuracy under strong noise conditions, this paper proposes a bearing fault diagnosis method based on multi-sensor data fusion. The main contributions of this paper are summarized as follows.

(1) A multi-scale quadratic convolutional neural network (MDQCNN) based on multi-sensor data fusion is proposed, which enhances the reliability and stability of the fault diagnosis system by fusing vibration signals collected from different sensors.
(2) A quadratic neuron embedded with an attention mechanism is introduced to replace the traditional convolution operation, improving the feature extraction capability of the traditional CNN and enabling the network to extract key features more efficiently, thereby enhancing fault diagnosis accuracy.
(3) To address the unique characteristics of bearing vibration signals, the first layer uses convolution kernels of different sizes to extract low, medium, and high-frequency features. Smaller convolution kernels are then applied for feature selection and extraction, with a cross attention mechanism used to capture more effective features, achieving more accurate fault diagnosis.
(4) Validation was conducted using bearing datasets from Case Western Reserve University, Politecnico di Torino and Shandong University of Science and Technology, with experimental results demonstrating that the proposed method delivers high diagnostic accuracy and robustness in noisy environments.

The rest of the paper is structured as follows: Sect. 2 introduces the relevant background knowledge of the proposed method. Section 3 details the proposed method and its specific implementation. Section 4 validates the effectiveness of the proposed method through experiments on three bearing datasets. Section 5 summarizes the research work presented in this paper.

2 Methodologies

2.1 Quadratic convolution

In the biological nervous system, different types of neurons interact and collectively perform various neural activities. Inspired by this, Fan et al. [40] proposed a novel type of neuron called a quadratic neuron. Quadratic neurons possess strong expressive power and are capable of more effectively capturing nonlinear relationships in sequence data. By replacing the inner product operation in traditional neurons with quadratic operations, a single quadratic neuron can not only perform XOR (exclusive OR) logic operations but also execute generalized fuzzy operations [41]. Figure 1 illustrates the internal operation mechanism of a quadratic neuron, mathematically expressed as Eq. (1):

$$\begin{aligned} f(x) &= \left(\sum_{i=1}^{n} \omega_i^r x_i + b^r\right)\left(\sum_{i=1}^{n} \omega_i^g x_i + b^g\right) \\ &\quad + \sum_{i=1}^{n} \omega_i^b x_i^2 + c \\ &= \left(W^r x^T + b^r\right)\left(W^g x^T + b^g\right) \\ &\quad + W^b (x \odot x)^T + c \end{aligned} \tag{1}$$

Among them, $\odot$ represents the Hadamard product, while W^r, W^g, W^b denote the weight vector, The terms $\omega^r, \omega^g, \omega^b \in R$ represents the weights, and $b_r, b_g, c \in R$ represents the bias, The indices r, g, and b are simply distinguishing markers without any special significance.

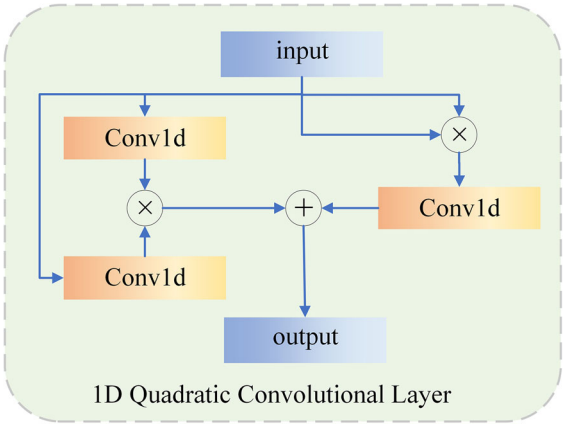


Fig. 1 The computational structure of quadratic neurons

Liao et al. [42] observed that when the constant term is removed, the quadratic neuron also incorporates an attention mechanism known as "qttention." By decomposing the learned activation function into a form similar to attention, its mathematical expression is given by Eq. (2):

$$\begin{aligned} f(x) &= (W^r x^T + b^r)(W^g x^T + b^g) + W^b (x \odot x)^T + c \\ &= x^T (\underbrace{x \odot W^b + W^g x^T W^r}_{qttention} \\ &\quad + \underbrace{W^g b^r + W^r b^g}_{bias}) \end{aligned} \tag{2}$$

Quadratic networks offer several advantages both theoretically and practically. First, quadratic neurons incorporate power functions in their calculations, enabling additional nonlinear mappings and stronger representational power, while traditional neurons can only achieve nonlinear mappings through activation functions. Second, quadratic neurons are piecewise polynomial functions, whereas traditional neurons are piecewise linear functions. Polynomials are better suited for modeling complex functions compared to linear functions. Finally, the outstanding performance of quadratic networks in bearing fault diagnosis under challenging conditions, such as strong noise and data imbalance, highlights their practicality and robustness in fault diagnosis.

2.2 Time frequency domain filter

The time domain filter primarily consists of a quadratic convolutional neural network, which is structured symmetrically with two quadratic convolutional layers. The first layer maps the input into 16 channels, while the second layer combines these 16 channels into a single output, effectively extracting time domain periodic pulses. In the time domain filter, kurtosis is employed as the objective function, and its loss calculation is as follows:

$$L_t = -\frac{\sum_{n=1}^{N} y(n)^4}{\left(\sum_{n=1}^{N} y(n)^2\right)^2} \tag{3}$$

In this case, (n) represents the output of the filter, and its length is equal to the input length, denoted as N.

The frequency domain filter is composed of fully connected neural networks, which process signals in the frequency domain through the Fast Fourier Transform (FFT), thereby enhancing the filtering ability of signal frequency components. Subsequently, the linear neural layer filters the frequency domain of the signal and performs Inverse Fast Fourier Transform (IFFT) to recover the time domain signal. Denote the signal passing through the time domain filter as $\hat{x}(t)$, the FFT $F(\cdot)$ is applied to convert the signal into the frequency domain.

$$\hat{X}(f) = \mathcal{F}\left(\hat{x}(t)\right) \tag{4}$$

According to the convolution theorem, the convolution of two time domain data is equivalent to the inner product of their Fourier transform domains. Frequency filter uses linear operations to filter signals in the frequency domain.

$$\hat{Y}(f) = \sum_{f=1}^{N} \omega_f(f)\hat{X}(f) + b_f \tag{5}$$

ωf and bf represent the weight and bias in the frequency domain, respectively, while $\hat{Y}(f)$ represents the filtered signal in the frequency domain.

Subsequently, the Inverse Fast Fourier Transform (IFFT), denoted as F^{-1}, is applied to recover the signal in the time domain.

$$\hat{y}(t) = \mathcal{F}^{-1}\left(\hat{Y}(f)\right) \tag{6}$$

The composition structure of the time frequency domain filter is shown in Fig. 2. The time frequency domain filter is composed of a cascaded time domain filter and a frequency domain filter. The introduction of quadratic convolution enhances the filter's extraction capability, while the linear neural filter enables the filtering of signals in the frequency domain [43].

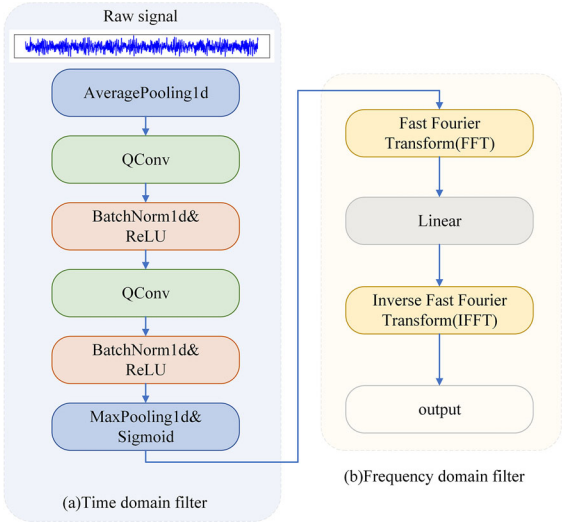


Fig. 2 Time frequency domain filter. **a** Time domain filter. **b** Frequency domain filter

2.3 Cross attention mechanism

As shown in Fig. 3, first, perform three convolutional transformations on the input features to obtain q, k, and v, respectively. These transformations help to separate different aspects of input features, thereby enhancing processing power. Subsequently, matrix multiplication is performed between q and k to calculate attention scores that quantify the correlation between different input elements. Then standardize the generated attention scores to ensure stability and consistency between different input sequences. Finally, multiply the standardized attention score by v to generate the output of attention. The specific process is described as follows.

$$\begin{aligned}\text{Cross_attention}(Q, K, V) &= A \\ &= \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot V\end{aligned} \tag{7}$$

Among them, A represents the output of attention, and $\sqrt{d_k}$ is the dimension of the input. The above describes the operation of cross attention between two inputs, and ultimately, all attention outputs are aggregated to obtain the final result.

The self attention mechanism is similar to the cross attention mechanism but differs in its scope. While the self attention mechanism computes the correlation between elements within the same input sequence

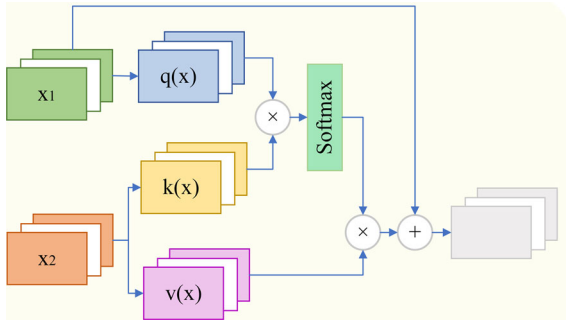


Fig. 3 Cross attention mechanism

to capture global dependencies of input features, the cross attention mechanism used in this paper focuses on integrating information across multiple inputs, thereby enhancing the effectiveness of overall feature fusion.

2.4 Information fusion strategy

The information provided by a single sensor is often limited, but information fusion methods can effectively leverage the complementary information from multiple sensors, thereby improving the accuracy of fault identification. Multi-sensor information fusion methods can be broadly categorized into three types: data level fusion, feature level fusion, and decision level fusion [2].

Feature level fusion relies on signal processing techniques to extract valuable features from raw sensor signals. While this approach enhances diagnostic capability, it often involves complex signal processing steps and may unintentionally lose critical information during the fusion process, potentially affecting classifier performance.Decision level fusion integrates diagnostic results from different sensor signals using predefined rules to improve recognition performance. However, training multiple classifiers requires significant computational resources. As shown in Fig. 4, data level fusion operates directly on raw data, maximizing the retention of fault signal characteristics. Among the three fusion methods, it typically achieves the best recognition performance. In data level fusion, multi-channel input is a common strategy. In this paper, vibration signals from bearings at different locations are combined into multi-channel data and fed into the model, effectively preserving both the temporal and spatial information of the original signals. This

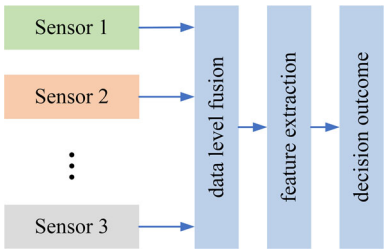


Fig. 4 Information fusion strategy

approach extracts cross-channel features from both the time and frequency domains, enhancing the accuracy of fault diagnosis.

3 MDQCNN model

3.1 Overall architecture and parameter configuration

The architecture proposed in this paper consists of four stages: data input, time frequency domain filtering, multi-scale feature extraction, and fault classification, as shown in Fig. 5. First, the signals collected by sensors at different positions contain the operating characteristics of the bearings at various locations. Using multi-channel data from sensors at different positions as input provides richer and more accurate signal information. The multi-channel data is then fed into the time frequency domain filtering module, where the time domain features of the signals are extracted. Fourier transform is applied to obtain the frequency domain representation of the signals, and inverse Fourier transform is used to obtain the filtered signals in the frequency domain. This process enriches the time frequency domain features of the signal, making it more suitable for feature extraction and classification in fault diagnosis tasks. Next, the extracted signals are input into convolutional layers with different kernel sizes to capture low-frequency, mid-frequency, and high-frequency features, capturing information at different scales. Small convolution kernels are used to further extract key features. Finally, a cross attention mechanism is applied to fuse the information from different scales, enhancing the model's understanding of temporal signals and cross-scale information. This enables the model to more accurately capture effective information, thereby improving the accuracy of fault diagnosis.

The parameter configuration of the architecture is shown in Table 1, where QConv, Pool, and FC represent the quadratic convolutional layer, MaxPool layer, and fully connected layer, respectively. In the QConv layer, the symbol "16-16 $\times$ 1" indicates that the convolutional layer contains 16 convolution kernels, each with a size of 16 $\times$ 1. "Pad" refers to the zero-padding operation. "BN" represents batch normalization, and "LeakyReLU" is the activation function. "Neuron 100" indicates that the fully connected layer contains 100 neurons.

3.2 Method process

This paper proposes a multi-scale quadratic convolutional neural network (MDQCNN) for intelligent fault diagnosis of bearings using multi-sensor data fusion. The framework of the proposed method is shown in Fig. 6. The specific operational steps are summarized as follows:

Step 1: Use signal acquisition equipment to collect multi-sensor datasets and obtain vibration signals of bearings at different positions.

Step 2: Divide the collected bearing vibration signals into training, validation, and testing sets.

Step 3: Apply time frequency domain filters to the input signal to reduce noise interference and improve signal quality.

Step 4: Use convolution kernels of different sizes to extract multi-scale features, and employ a cross attention mechanism to extract relevant features, ultimately outputting fault diagnosis results.

Step 5: Visualize and analyze the diagnostic results using confusion matrices and t-SNE dimensionality reduction techniques [44] to display the model's diagnostic performance for visual evaluation and analysis.

4 Experiment

4.1 Experimental configuration

4.1.1 Comparing models and training settings

In order to comprehensively evaluate the diagnostic performance of the proposed model in complex working conditions, we selected several state-of-the-art bearing fault diagnosis methods as baseline comparison

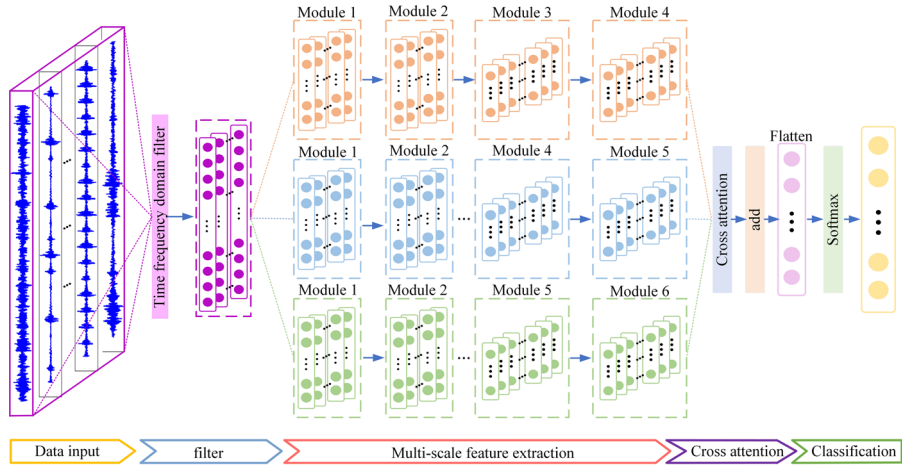


Fig. 5 Structure of the proposed diagnostic model

Table 1 Parameter configuration of MDQCNN

Network module	Layer	Scale	Parameter
Module1	QConv 1	1	Kernel 16-16×1, Stride 8, Pad 1, BN, LeakyReLU
		2	Kernel 16-32×1, Stride 4, Pad 1, BN, LeakyReLU
		3	Kernel 16-64×1, Stride 2, Pad 1, BN, LeakyReLU
	Pool 1	1-3	MaxPool1d(2,2)
Module 2	QConv 2	1-3	Kernel 32-3×1, Stride 1, Pad 1, BN, LeakyReLU
	Pool 2	1-3	MaxPool1d(2,2)
Module 3	Qconv 3	1-3	Kernel 64-3×1, Stride 1, Pad 1, BN, LeakyReLU
	Pool 3	1-3	MaxPool1d(2,2)
Module 4	Qconv 4	1-3	Kernel 64-3×1, Stride 1, Pad 1, BN, LeakyReLU
	Pool 4	1-3	MaxPool1d(2,2)
Module 5	QConv 5	2-3	Kernel 64-3×1, Stride 1, Pad 1, BN, LeakyReLU
	Pool 5	2-3	MaxPool1d(2,2)
Module 6	QConv 6	3	Kernel 64-3×1, Stride 1, Pad 1, BN, LeakyReLU
	Pool 6	3	MaxPool1d(2,2)
Module 7			Cross attention
FC	Fc 1		Neuron 100, LeakyReLU, Dropout(0.5)
	Fc 2		Neuron 9, Softplus
Classifier			Softmax

models, including WDCNN [25], QCNN [42], DRSN [45], EWSNet [46], DCA-BiGRU [47], and TFN [48]. WDCNN utilizes a wide convolutional kernel as the first layer and is a classic CNN model designed specifically for bearing fault diagnosis. QCNN, by introducing quadratic convolution instead of traditional convolution, demonstrates strong noise resistance and performs exceptionally well in fault diagnosis. DRSN incorporates a residual shrinkage module into traditional deep residual networks, enhancing its performance. EWSNet leverages a large convolution kernel in the first layer to extract features and performs multi-scale feature fusion. DCA-BiGRU excels at extracting both high and low-frequency features from signals. TFN is a neural network model designed for intelligent fault diagnosis, embedding time frequency transforms and offering strong interpretability.

To increase the sample size, data augmentation was performed on the original data using a window size of 2048 and a sliding step size of 1024, generating 500

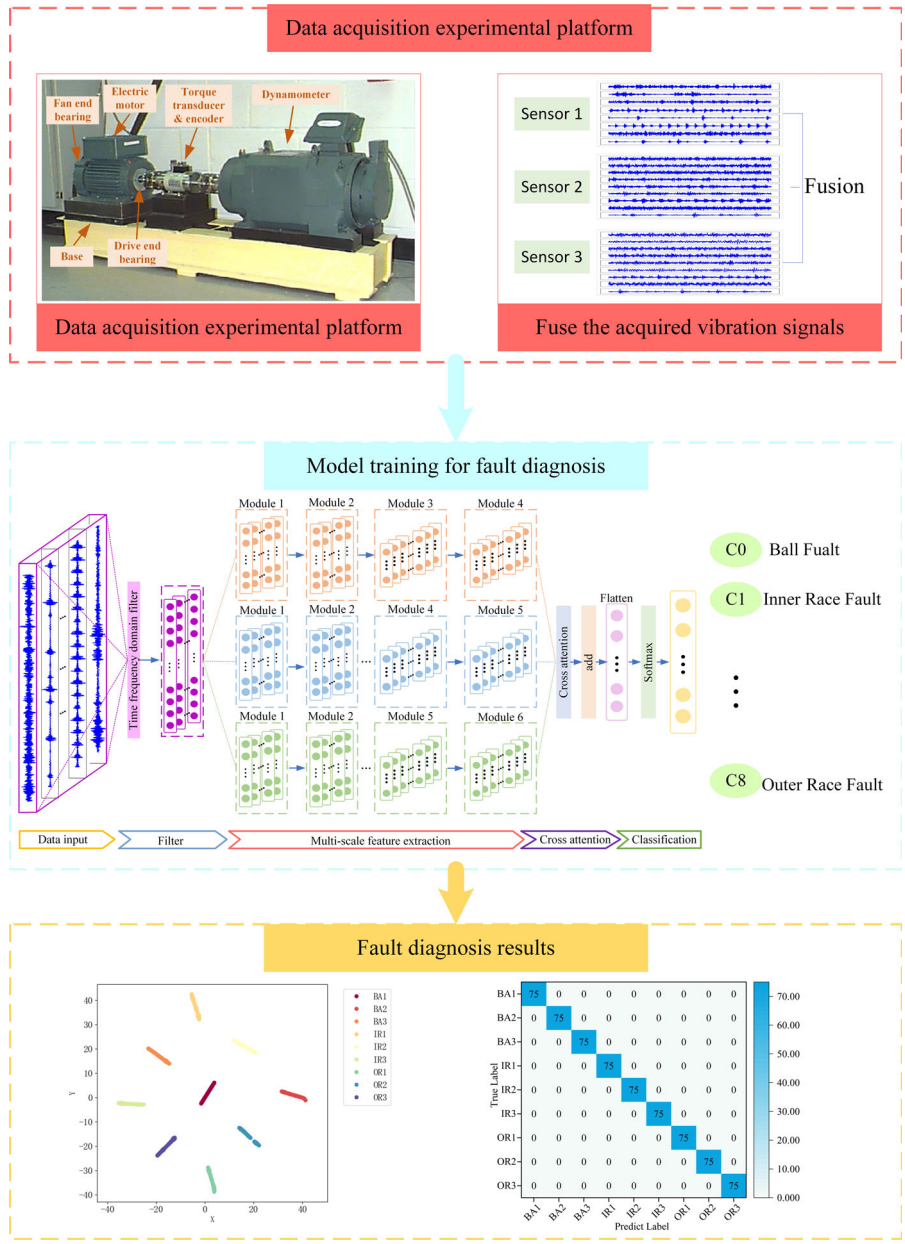


Fig. 6 The flowchart of the fault diagnosis method proposed

samples for different fault types. The data was then split into training, validation, and test sets in a 7:1.5:1.5 ratio. The model was trained using the Adam optimizer with the LSR loss function, a batch size of 100 epochs, and a learning rate set to 0.0001. It is important to note that, to ensure fairness in the comparative experiment, all baseline models were trained using the learning rate corresponding to their best diagnostic performance under identical conditions.

The experiment was conducted on a Windows 11 system computer, with hardware configuration including Intel processor, 8GB of RAM, and NVIDIA GeForce RTX 4060 graphics card. Using the Python programming language and implementing it on the Pycharm platform through the PyTorch deep learning framework.

4.1.2 Signal preprocessing

To make the collected data as representative as possible of samples obtained under actual working conditions, this paper introduces Gaussian white noise with varying Signal-to-Noise Ratios (SNR) to simulate noisy samples typically encountered in real-world environments [49]. The method for calculating the SNR is shown in Eq. (8).

$$\mathrm{SNR} = 10\log_{10}\left(\frac{P_s}{P_n}\right) \tag{8}$$

Among them, SNR represents the strength of the signal relative to the noise, P_s is the signal power, and P_n is the noise power. A lower SNR indicates a higher level of noise.

4.1.3 Evaluation indicators

We evaluate the performance of all considered methods using two commonly used metrics: the False Positive Rate (FPR) and the F1 score. The definitions of these two indicators are as follows:

$$\mathrm{FPR} = \frac{FP}{FP + TN} \tag{9}$$

$$\mathrm{Recall} = \frac{TP}{TP + FN} \tag{10}$$

$$\mathrm{Precision} = \frac{TP}{TP + FP} \tag{11}$$

$$F1\ \mathrm{score} = \frac{2 \cdot \mathrm{Precision} \cdot \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} = \frac{2 \cdot TP}{2TP + FN + FP} \tag{12}$$

TP, TN, FP, and FN represent the number of true positives, true negatives, false positives, and false negatives, respectively.

4.2 Case 1 CWRU dataset

4.2.1 Dataset description

The CWRU bearing dataset is a widely used public dataset for fault diagnosis and health monitoring, often employed to evaluate model performance and has strong representativeness [47]. The data acquisition experimental platform is shown in Fig. 7, which includes vibration signals collected from bearings at the drive end, fan end, and base of the motor under four different loads. The corresponding speeds for each load are 1797 r/min (0 hp), 1772 r/min (1 hp), 1750 r/min (2 hp), and 1730 r/min (3 hp). Among them, the signal sampling frequency at the drive end is divided into 12 kHz and 48 kHz. By using electric spark technology, three different degrees of damage, 0.007 inches, 0.014 inches, and 0.021 inches, were implanted in the rolling elements, inner rings and outer rings of the bearing, resulting in nine different types of faults. The detailed information is provided in Table 2. The vibration signals under each health state are shown in Fig. 8. In this paper, experimental data from three sensors located at the drive end, fan end, and base under the 12 kHz condition were selected.

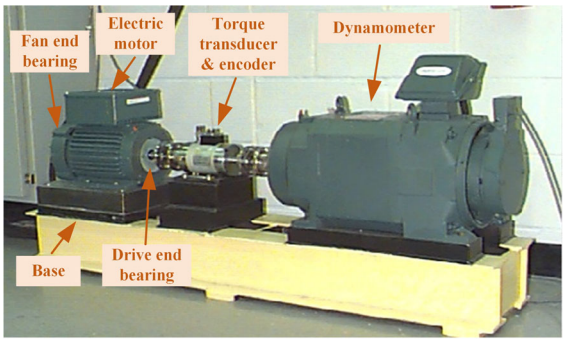


Fig. 7 CWRU data acquisition experimental platform

4.2.2 Experiments under different signals

To verify the effectiveness of multi-sensor information fusion in fault classification and recognition, we compared the performance of the fault diagnosis model in classifying the health status of bearings using different signals. Model A uses data from the drive end, Model B uses data from the fan end, Model C uses data from the base, Model D integrates data from the drive end and fan end, Model E integrates data from the drive end and base, and Model F integrates data from the fan end and base. To further demonstrate the advantages of quadratic convolution over traditional convolution, Model G replaced quadratic convolution with traditional convolution and conducted experiments on the fused datasets from the drive end, fan end, and base. All experiments were conducted in an environment with a signal-to-noise ratio of -2. To ensure the stability of the results, conduct five independent experiments for

Table 2 Detailed information of CWRU dataset

Fault type	Loads (hp)	Fault size (inch)	Label
Ball Fault	0/1/2/3	0.007	BA1
		0.014	BA2
		0.021	BA3
Inner Race Fault		0.007	IR1
		0.014	IR2
		0.021	IR3
Outer Race Fault		0.007	OR1
		0.014	OR2
		0.021	OR3

each method. The specific results are shown in Table 3, representing the average of the five experiments.

From the table, it can be seen that the average F1 score of Model E is the lowest, at only 93.90%. Model A ranks second, with an average F1 score of 95.21%, which is 1.31% higher than that of Model E. The average F1 scores of Models B, C, and D differ slightly, all exceeding 98.50%. The average F1 scores of Models F, G, and MDQCNN are relatively close, all surpassing 99.50%. Among these, MDQCNN achieves the highest average F1 score, reaching 99.88%.

Under the 0 hp load, the F1 score of Model A is the lowest, at only 91.12%. The F1 scores of both the G model and MDQCNN are the highest, each reaching 99.79%. Under the 1 hp load, the F1 score of Model E is the lowest, at just 84.66%. In contrast, MDQCNN achieves the highest F1 score of 99.97%, which is 15.31% higher than that of Model E. Under the 2 hp load, Model E still has the lowest F1 score, at 91.37%, while MDQCNN leads with the highest F1 score of 99.79%, which is 8.42% higher than that of Model E. Under the 3 hp load, Model A has the lowest F1 score of 97.31%, followed by Model B with an F1 score of 99.85%. All other models achieve 100% accuracy.

Figure 9 shows the comparison of F1 scores for various models under different loads. From the figure, it can be observed that although the Model E performs relatively well under the 0 hp and 3 hp loads, its diagnostic results fluctuate significantly across different loads. Specifically, under the 1 hp and 2 hp loads, the F1 score drops substantially, with the lowest F1 score occurring at 1 hp load. In contrast, although Model A has a lower F1 score under various loads, its overall performance remains relatively stable, with smaller fluctuations compared to Model E. The F1 scores of Models B, C, and D are closely aligned, showing consistent performance across the loads. It is noteworthy that the F1 score of the Model F is significantly higher than that of the Model G under the 1 hp load. MDQCNN stands out as the best-performing model across all loads, exhibiting the smallest variation in F1 scores, which demonstrates its exceptional stability.

In summary, when using the vibration signal from a single sensor for fault diagnosis, the F1 score is relatively low. However, fusing the vibration signals from two sensors improves diagnostic performance, though some instability may arise. By combining the vibration signals from three sensors, the F1 score reaches its highest value and becomes more stable, clearly demonstrating the advantages of multi-sensor information fusion in enhancing fault diagnosis performance.

4.2.3 Comparative experiment

To verify the effectiveness of the proposed fault diagnosis method and its robustness against noise interference, we compared the model with six state-of-the-art fault diagnosis models. A detailed introduction of these models and their respective parameter settings is provided in Section 4.1.1. The experiments were conducted under four loads: 0hp, 1hp, 2hp, and 3hp, with five different noise levels. The specific classification results are presented in Table 4, with the F1 score representing the average value of each model after five independent runs.

Based on the results in the table, the following key observations can be made. First, the fault diagnosis performance of the proposed model under various loads and noise interference levels generally outperforms the other comparative models. The average F1 score of the

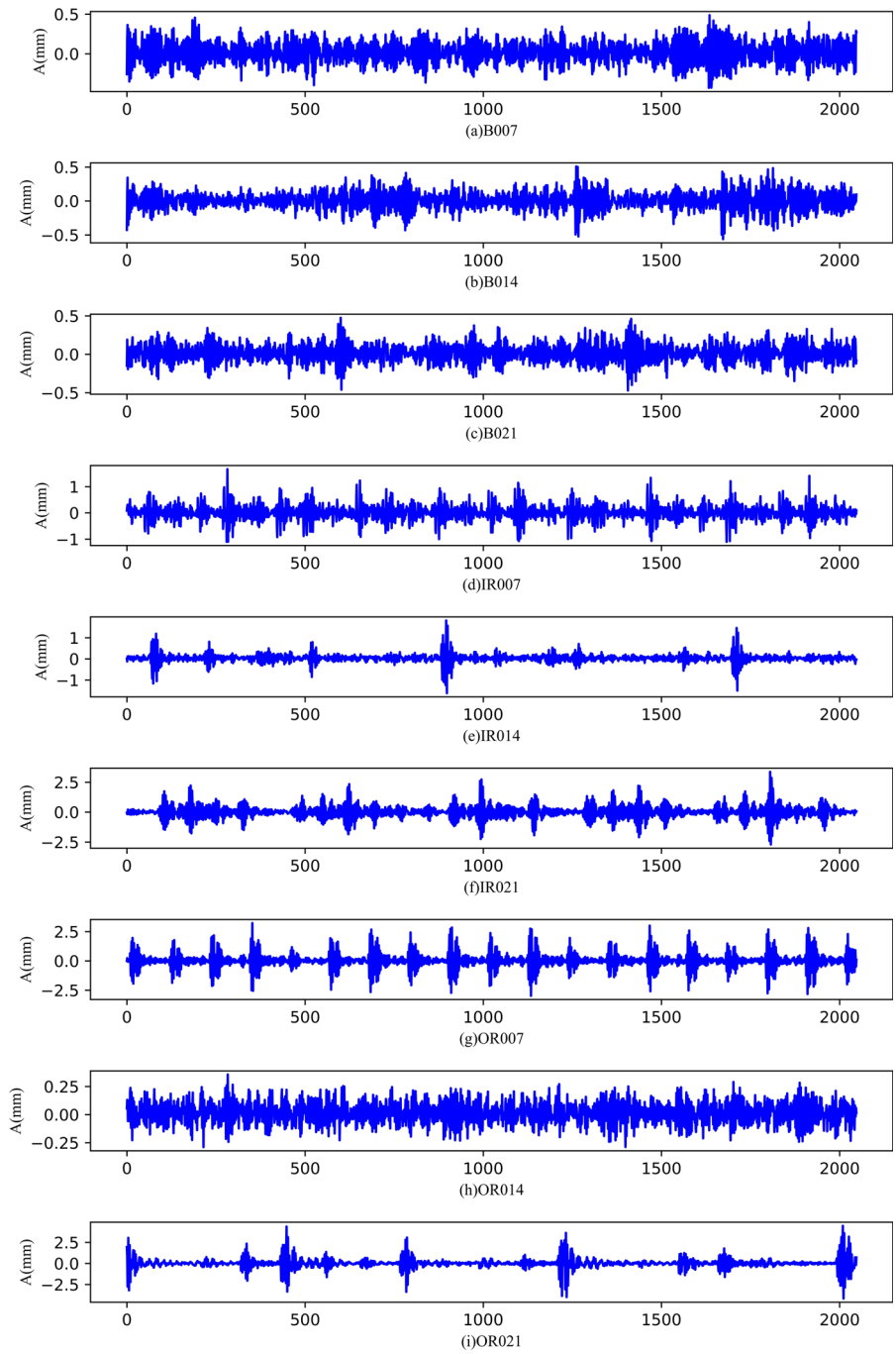


Fig. 8 Vibration signals under 9 health states in the CWRU dataset

proposed model exceeds 93.79% across all loads, with the most notable improvement under the 1hp load. In this condition, the proposed model's average F1 score is 9.18% higher than the second-best model and 12.49% higher than the third-best model. Although the proposed model may not achieve the highest F1 score in all cases, the difference from the highest score is minimal. Therefore, it can be concluded that the proposed model demonstrates significant competitiveness under different loads and noise interference scenarios.

Secondly, the proposed model demonstrates significant advantages under extreme noise interference con-

Table 3 F1 scores under different inputs and operating conditions

Model	0hp		1hp		2hp		3hp		Average	
	F1 (%)	FPR (%)	F1 (%)	FPR (%)	F1 (%)	FPR (%)	F1 (%)	FPR (%)	F1 (%)	FPR (%)
A	91.12	1.07	94.56	0.64	97.86	0.26	97.31	0.33	95.21	0.57
B	98.91	0.12	96.92	0.36	98.41	0.19	99.85	0.01	98.52	0.17
C	97.53	0.30	97.71	0.27	99.11	0.10	100	0.00	98.58	0.16
D	96.97	0.36	97.86	0.27	99.64	0.03	100	0.00	98.61	0.16
E	99.58	0.04	84.66	1.77	91.37	1.13	100	0.00	93.90	0.73
F	99.34	0.07	99.88	0.01	98.92	0.12	100	0.00	99.53	0.05
G	99.79	0.01	98.75	0.15	99.49	0.05	100	0.00	99.50	0.05
MDQCNN	99.79	0.01	99.97	0.01	99.79	0.02	100	0.00	99.88	0.01

ditions. In the case of SNR = −10, the F1 score of the proposed model outperforms all other compared models across all loads. Notably, in the 1hp load, its F1 score is 20.30% higher than that of the second-best model. Additionally, the F1 score of QCNN consistently surpasses that of WDCNN. Since both models share the same network structure, this result validates that quadratic neurons are a simple and effective method for enhancing network performance.

To further analyze the learning performance of the proposed model compared to other models, we used the t-SNE method [41] to visualize the features of the last convolutional output from the four models with the highest F1 scores at SNR = −10 under the 3hp load, as shown in Fig. 10. Different colors represent different types of bearing faults. MDQCNN successfully distinguished all five types of bearings clearly, while the other models achieved good classification results but still had some points of confusion.

Finally, to comprehensively evaluate the diagnostic performance of each model under noise interference conditions, a detailed analysis was conducted using a confusion matrix, as shown in Fig. 11. Most of the misclassifications occurred in samples related to ball faults, likely due to the fact that the characteristic frequency of ball faults is determined by the cage speed, both of which are influenced by random rotation. Therefore, the random interference introduced by strong noise increases the difficulty of identifying ball faults of varying degrees. Additionally, the EWSNet and TFN models faced significant challenges in identifying OR2. These issues stem from the limited feature extraction capabilities of their networks, which hinder their ability to capture OR2 features. It is worth noting that although the diagnostic accuracy of the proposed model on BA3 is lower than that of the EWSNet model, the EWSNet model accurately diagnosed only 7 samples of OR2. In comparison, the proposed model performs better in other categories, demonstrating its broader applicability and robustness.

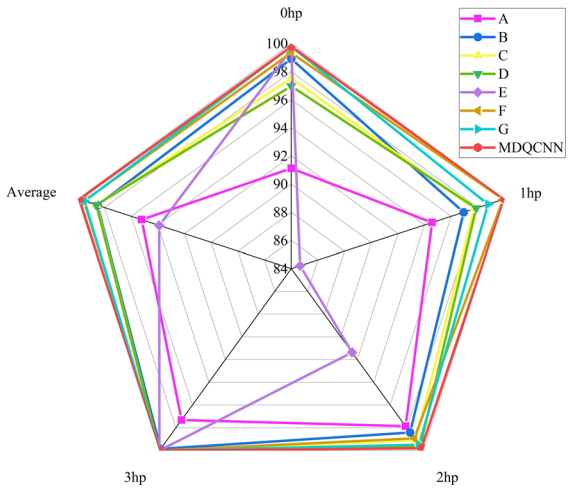


Fig. 9 Comparison of F1 scores of various models under different loads

4.2.4 Ablation experiment

To verify the effectiveness of the added modules, ablation experiments were conducted under a 0 hp load and five different noise environments. In these experiments, Model A has the same multi-scale feature extraction module structure as the proposed model but replaces quadratic convolution with traditional convolution, and the obtained multi-scale features are directly combined using the addition operation. Model B has the same

Table 4 Classification results of each model under different loads and noise conditions

Loads	Model	−2		−4		−6		−8		−10		Average	
		F1 (%)	FPR (%)	F1 (%)	FPR (%)	F1 (%)	FPR (%)	F1 (%)	FPR (%)	F1 (%)	FPR (%)	F1 (%)	FPR (%)
0hp	WDCNN	99.25	0.08	97.49	0.30	92.85	0.87	77.99	2.60	62.95	4.42	86.10	1.65
	QCNN	99.25	0.08	95.97	0.49	90.27	1.22	76.68	2.78	63.32	4.40	85.09	1.79
	DRSN	99.34	0.07	92.69	0.84	78.48	2.40	64.42	3.96	55.04	5.18	77.99	2.49
	EWSNet	100	0.00	99.97	0.01	96.44	0.41	82.32	1.81	72.26	2.93	90.19	1.03
	DCA-BIGRU	99.11	0.10	96.91	0.37	88.88	1.36	82.83	2.10	71.17	3.52	87.78	1.49
	TFN	95.65	0.53	90.82	1.14	83.06	2.05	75.25	3.07	66.69	4.00	82.29	2.15
	MDQCNN	99.79	0.01	99.52	0.05	97.32	0.32	96.22	0.47	90.34	1.23	96.63	0.41
1hp	WDCNN	94.33	0.66	87.27	1.44	81.05	2.22	72.33	3.38	57.56	5.26	78.50	2.59
	QCNN	95.52	0.53	90.85	1.10	82.31	2.13	73.50	3.30	60.43	4.80	80.52	2.37
	DRSN	91.92	1.01	85.81	1.71	72.26	3.19	60.24	4.70	51.72	5.41	72.39	3.20
	EWSNet	99.04	0.11	86.44	1.31	78.27	2.43	78.14	2.29	64.61	3.55	81.30	1.93
	DCA-BIGRU	98.03	0.24	94.88	0.61	87.67	1.43	76.73	2.82	65.74	4.06	84.61	1.83
	TFN	93.31	0.82	85.70	1.70	79.94	2.27	69.85	3.31	59.02	4.83	77.56	2.58
	MDQCNN	99.97	0.01	98.44	0.18	95.88	0.51	88.63	1.18	86.04	1.73	93.79	0.72
2hp	WDCNN	99.16	0.09	96.70	0.40	84.08	1.89	71.98	3.45	60.84	4.84	82.55	2.13
	QCNN	98.37	0.19	91.90	0.99	85.63	1.72	67.93	3.74	61.61	4.64	81.08	2.25
	DRSN	93.42	0.83	77.65	2.37	70.84	3.24	61.56	4.41	55.83	5.08	71.86	3.18
	EWSNet	100	0.00	99.19	0.09	88.97	1.34	71.73	2.88	59.76	4.44	83.93	1.75
	DCA-BIGRU	99.58	0.04	96.78	0.39	90.60	1.16	78.91	2.59	67.14	4.02	86.60	1.64
	TFN	94.48	0.65	87.09	1.52	77.21	2.58	72.11	3.26	61.84	4.41	78.54	2.48
	MDQCNN	99.79	0.02	99.64	0.03	96.20	0.47	93.42	0.81	83.10	2.04	94.43	0.67
3hp	WDCNN	100	0.00	99.82	0.01	99.55	0.05	96.92	0.38	87.20	1.52	96.69	0.39
	QCNN	99.61	0.04	99.05	0.11	99.25	0.08	96.34	0.45	92.60	0.91	97.37	0.31
	DRSN	85.87	1.41	99.58	0.04	96.70	0.40	89.47	1.25	70.85	3.42	88.49	1.30
	EWSNet	100	0.00	100	0.00	99.85	0.01	99.55	0.05	86.23	1.40	97.12	0.29
	DCA-BIGRU	99.97	0.01	99.97	0.01	99.19	0.09	95.37	0.57	87.43	1.53	96.38	0.44
	TFN	99.07	0.11	96.67	0.41	88.14	1.43	82.47	2.07	73.11	3.19	87.89	1.44
	MDQCNN	100	0.00	100	0.00	99.88	0.01	99.31	0.08	97.20	0.34	99.27	0.08

structure as the proposed model but does not perform time frequency domain filtering. Models C, D, and E share the same structure as the proposed model but use single-channel input, respectively taking vibration signals from sensors at the drive end, fan end, and base. Each model underwent five independent experiments under each of the five noise environments, and the average results are shown in Fig. 12.

As shown in Fig. 12, Model A performs the worst when the SNR is −10. Model B achieves higher accuracy than Model A, with the gap between them decreasing as the noise level decreases. The proposed model achieves the highest accuracy, with accuracy exceeding 90% across all five noise environments. A comparison of the accuracy between Model A and the proposed model across the five noise environments demonstrates that time frequency domain filtering significantly impacts model accuracy. In high-noise environments, time frequency domain filtering effectively enhances the model's fault diagnosis accuracy.

Comparing the accuracy of Model A and Model B across the five noise environments highlights the effectiveness of using quadratic convolution combined with the cross attention mechanism in the multi-scale feature extraction stage for fault diagnosis.

Among Models C, D, and E, Model D, which uses single-channel fan end signal input, achieves the highest accuracy across the five noise environments, while

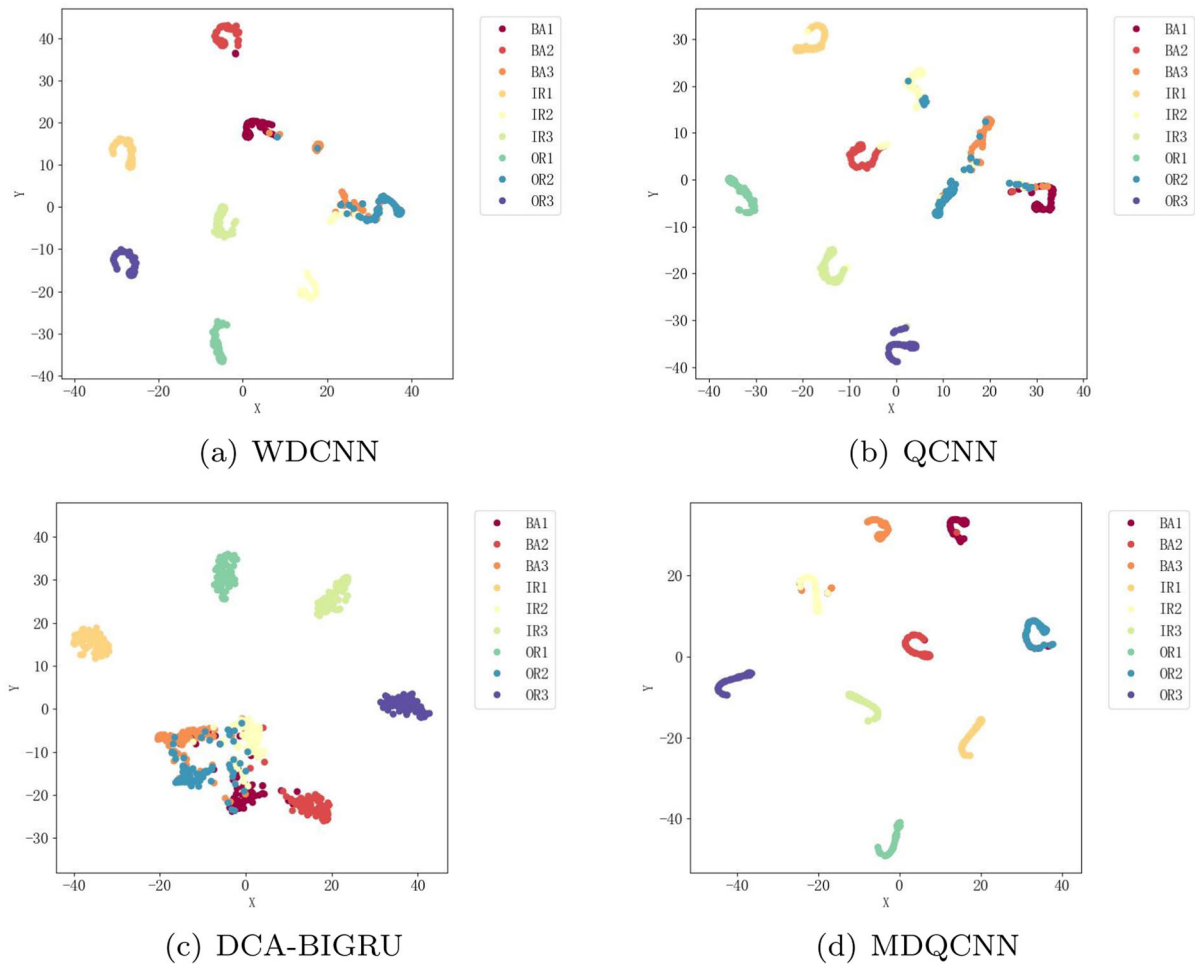


Fig. 10 t-SNE visualization under CWRU dataset

Model E, which uses single-channel drive end signal input, shows the lowest accuracy. This indicates that vibration signals from sensors at different locations can impact the model's fault diagnosis accuracy. The proposed model achieves high accuracy across all five noise environments, suggesting that using vibration signals from different sensor locations as multi-channel input can effectively enhance the model's accuracy in fault diagnosis.

4.3 Case 2 The aerospace bearing dataset

4.3.1 Dataset description

The experimental data used in this section comes from the simulation test bench for aircraft engine bearing faults at the Politecnico di Torino in Italy. The experimental setup is shown in Fig. 13. This test bench is capable of measuring the vibration signals of aviation bearings under different high-speed and heavy load conditions. The experimental data includes 7 states: normal condition (NC), minor inner ring failure (IF1), inner ring failure (IF2), severe inner ring failure (IF3) minor rolling element failure (RF1), rolling element failure (RF2), and severe rolling element failure (RF3). These states correspond to specific damage diameters, as shown in Table 5, specifically 150 μm, 250 μm, and 450 μum. The signal sampling frequency is 51.2 kHz. In order to simulate the high load operation of aircraft engine bearings under real working conditions, this study selected vibration data from positions A1 and A2 at a speed of 30,000 r/min and a load of 1000 N to construct a dataset. The acceleration signals of each state are collected through three channels: X, Y, and Z.

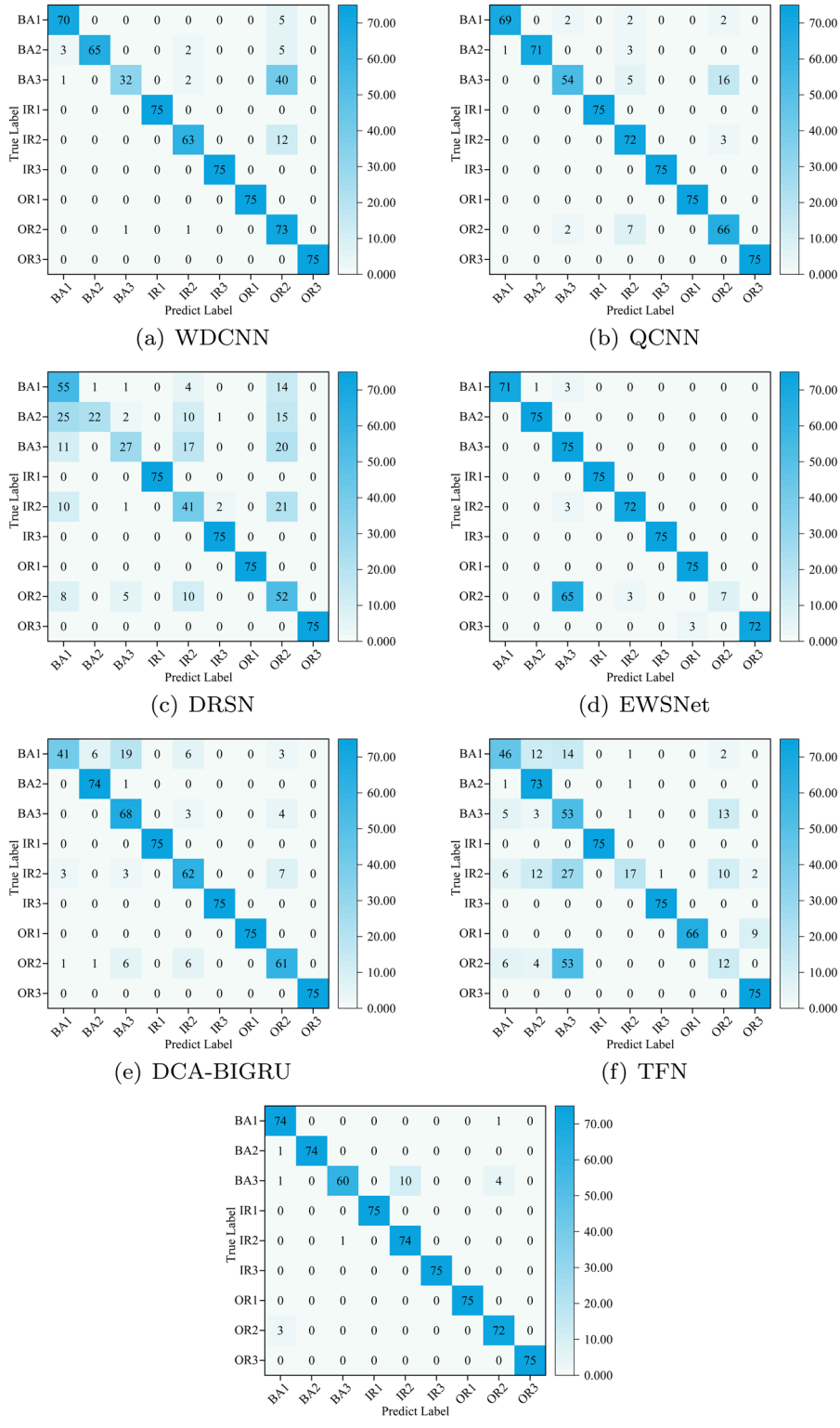


Fig. 11 Confusion matrices of various models in the CWRU dataset

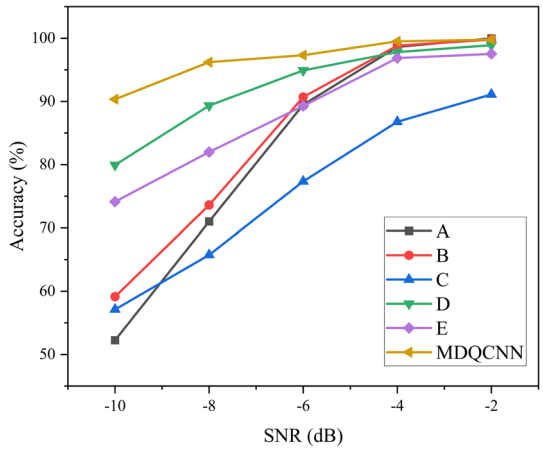


Fig. 12 Experiments on ablation under five types of noise environments

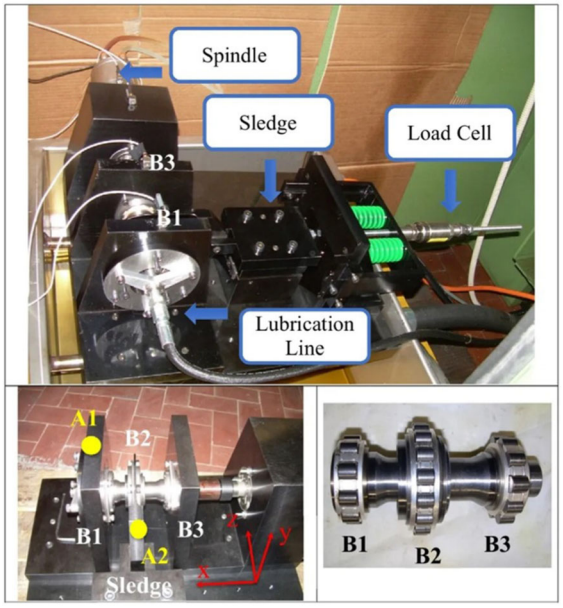


Fig. 13 The aerospace bearing data acquisition platform

4.3.2 Experiments under different signals

To evaluate the effectiveness of multi-sensor information fusion in fault classification and recognition, this study compared the classification performance of various fault diagnosis models on different datasets. Specifically, Model A performs experiments using data from Sensor 1, Model B uses data from Sensor 2, and Model C replaces quadratic convolution with traditional convolution and conducts experiments after fusing the signals from both sensors. All experiments were carried out in an environment with a signal-to-noise ratio of −2. Table 6 presents the F1 score results from five independent runs.

From Table 6, it is evident that the proposed model achieves higher F1 scores than the other models across all five independent experiments. The average F1 score of the proposed model reaches 100%, which is 0.08% higher than that of the second-highest model, Model C. In contrast, the average F1 scores of Model A and Model B are significantly lower across the five experiments. These results highlight that multi-sensor data fusion can significantly enhance the performance of fault diagnosis.

Figure 14 shows the F1 score comparison results of each model after five independent runs. Among them, Model A and Model C showed a significant difference in F1 scores between the highest and lowest in the five experiments, indicating significant instability. In contrast, the F1 score of the proposed model consistently remains at a high level and fluctuates less, demonstrating higher stability. Although the C model also achieved high F1 scores in the last four experiments, its F1 score in the first experiment was relatively low and showed relative instability. In summary, the introduction of quadratic convolution combined with multi-sensor information fusion effectively improves the performance and stability of the model.

4.3.3 Comparative experiment

To evaluate the effectiveness of the proposed fault diagnosis method and its noise resistance capabilities, this paper compares the model with six state-of-the-art fault diagnosis models. Detailed descriptions and parameter settings for each model are provided in Section 4.1.1. The experiments were conducted under five different noise levels, and the F1 scores presented in Table 7 represent the averages from five independent runs.

Based on the results in Table 7, several key conclusions can be drawn. Firstly, the proposed model achieves the highest average F1 score among all compared models, surpassing the second-highest WDCNN by 3.34% and the third-highest DCA-BIGRU by 4.15%. Across all noise conditions, the F1 score of the proposed model consistently exceeds 98.81%. Although the proposed model may not achieve the

Table 5 Detailed information on the bearing dataset of Politecnico di Torino

Fault size	Speed (r/min)/Load (N)	Fault type	Label
	30,000/1000	Normal	NC
150		Inner Ring Fault	IF1
250			IF2
450			IF3
150		Roller Fault	RF1
250			RF2
450			RF3

Table 6 Five classification results under different inputs

Model	1		2		3		4		5		Average	
	F1 (%)	FPR (%)	F1 (%)	FPR (%)	F1 (%)	FPR (%)	F1 (%)	FPR (%)	F1 (%)	FPR (%)	F1 (%)	FPR (%)
A	99.61	0.06	100	0.00	100	0.00	99.62	0.06	99.61	0.06	99.76	0.03
B	99.80	0.03	99.61	0.06	99.80	0.03	99.80	0.03	99.80	0.03	99.76	0.03
C	99.61	0.06	100	0.00	100	0.00	100	0.00	100	0.00	99.92	0.01
MDQCNN	100	0.00	100	0.00	100	0.00	100	0.00	100	0.00	100	0.00

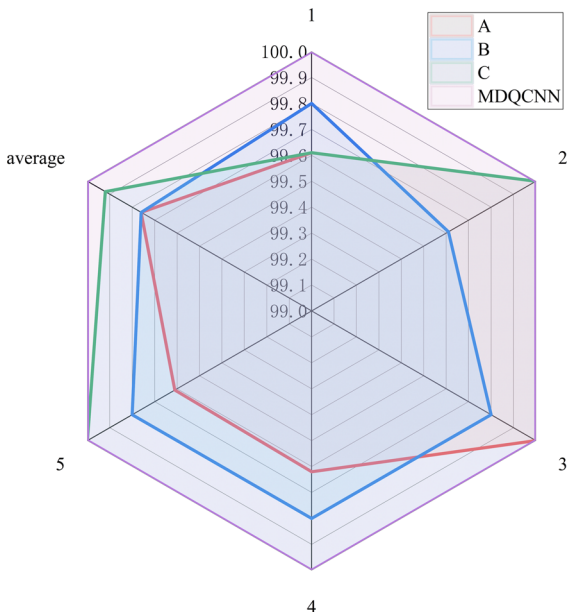


Fig. 14 F1 scores of five experiments on the bearing dataset of Politecnico di Torino

highest F1 score under certain conditions, the difference from the top-performing model is minimal. This demonstrates that the proposed model exhibits robust competitiveness even under noisy conditions.

Secondly, the proposed model demonstrates significant advantages under severe noise conditions. When the SNR is −10, the F1 score of DRSN is the lowest, at only 45.60%, while the proposed model achieves the highest F1 score of 98.81%, surpassing the second-highest WDCNN by 11.60%. To further analyze the performance of the models, the features output by the last convolutional layer of each model were visualized using t-SNE. As shown in Fig. 15, different colors represent different bearing health conditions. The results indicate that the proposed model accurately classifies all seven types of bearings.

Additionally, F1 scores were calculated for each model across different fault types. As shown in Table 8, DRSN, EWSNet, and TFN exhibit low recognition accuracy for NC and RF2, while WDCNN, QCNN, and DCA-BIGRU also show classification errors for NC and IF2. In contrast, the proposed model demonstrates superior performance, with only a small number of misclassifications in IF3 and RF2. For all other fault categories, the recognition accuracy reaches 100%, further validating the effectiveness of the proposed method.

Table 7 Classification results of various models under different noise conditions

Model	−2		−4		−6		−8		−10		Average	
	F1 (%)	FPR (%)	F1 (%)	FPR (%)	F1 (%)	FPR (%)	F1 (%)	FPR (%)	F1 (%)	FPR (%)	F1 (%)	FPR (%)
WDCNN	99.92	0.01	99.73	0.04	98.95	0.16	95.14	0.79	87.21	2.13	96.19	0.62
QCNN	98.10	0.31	99.08	0.14	95.71	0.72	92.13	1.31	79.91	3.27	92.98	1.15
DRSN	99.96	0.01	98.67	0.21	82.73	2.91	59.52	6.59	45.60	8.93	77.29	3.73
EWSNet	100	0.00	100	0.00	100	0.00	96.31	0.59	70.76	4.22	93.41	0.96
DCA-BIGRU	99.30	0.11	99.53	0.07	98.32	0.27	95.82	0.68	83.97	2.61	95.38	0.74
TFN	96.16	0.62	91.05	1.47	84.72	2.52	62.86	5.79	49.40	8.07	76.83	3.69
MDQCNN	100	0.00	100	0.00	99.88	0.01	98.96	0.16	98.81	0.19	99.53	0.07

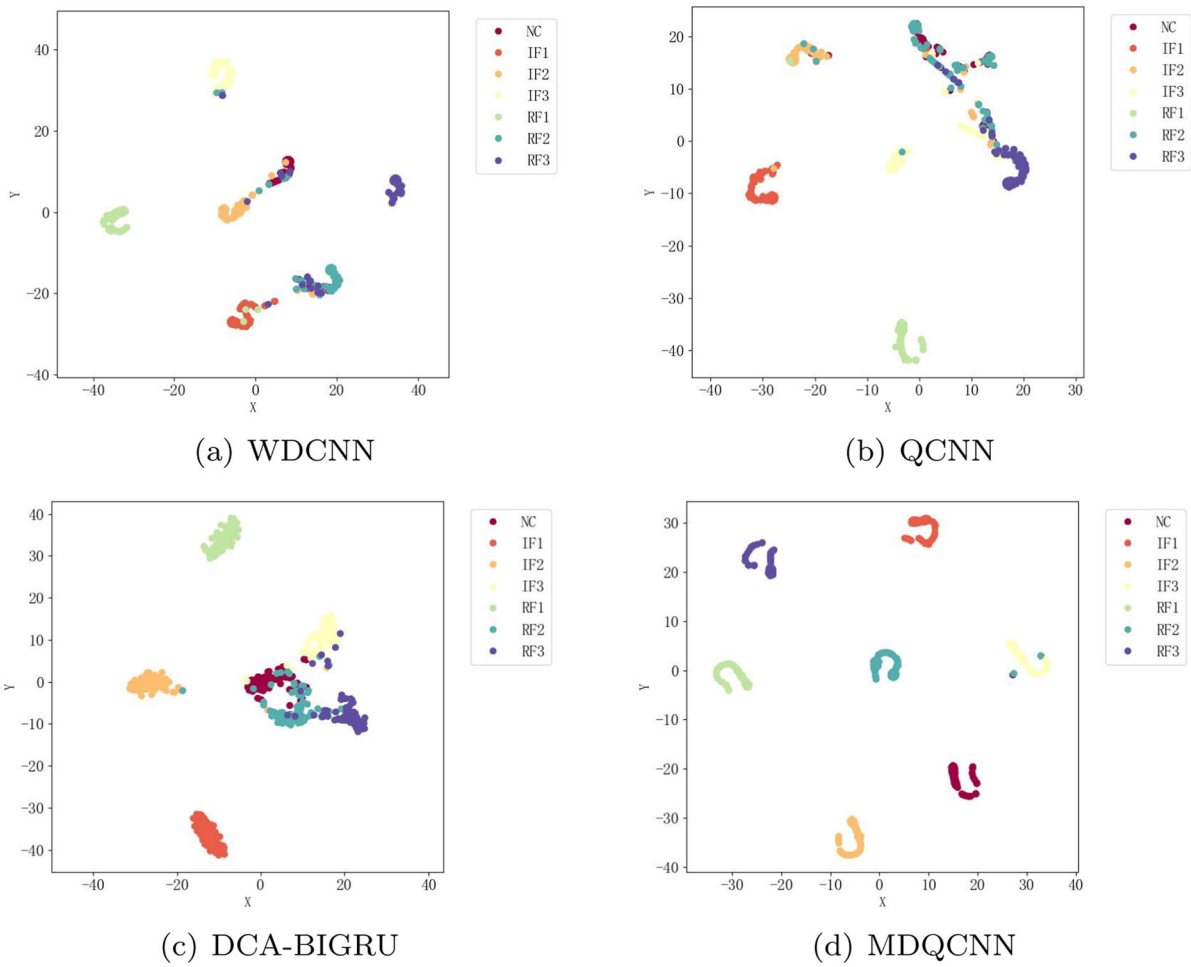


Fig. 15 t-SNE visualization under the bearing dataset of Politecnico di Torino

Table 8 F1 scores of each model in different states

Model	F-measure of various conditions NC (%)	IF1 (%)	IF2 (%)	IF3 (%)	RF1 (%)	RF2 (%)	RF3 (%)
WDCNN	65.33	90.67	85.33	100	96.00	90.67	98.67
QCNN	72.00	93.33	84.00	98.67	98.67	40.00	88.00
DRSN	36.00	97.33	61.33	62.67	100	16.00	0.00
EWSNet	90.67	89.33	100	98.67	100	0.00	24.00
DCA-BIGRU	84.00	100	80.00	94.67	100	78.67	80.00
TFN	52.00	82.67	92.00	24.00	68.00	6.67	6.67
MDQCNN	100	100	100	98.67	100	96.00	100

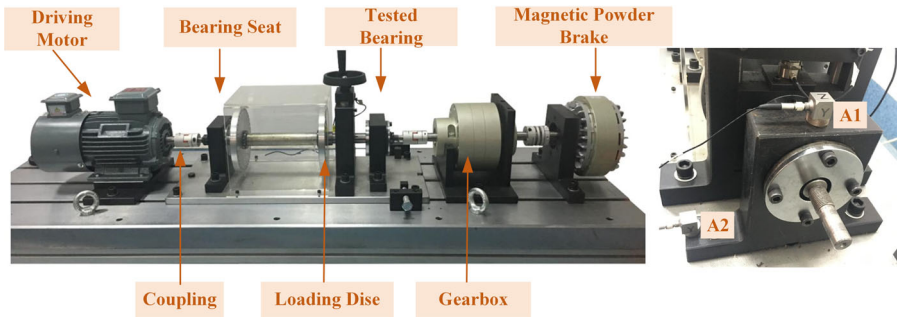


Fig. 16 SDUST Data acquisition platform

4.4 Case 3 The bearing dataset of Shandong University of Science and Technology

4.4.1 Dataset description

The data used in this section is from the bearing dataset of Shandong University of Science and Technology [50]. Figure 16 shows the data acquisition platform, which consists of a drive motor, two bearing housings, two loading discs, three couplings, a gearbox, and a magnetic powder brake. The bearing under test is located on the second bearing housing. The tested bearing has ten health conditions: NC, IF0.2, IF0.4, IF0.6, OF0.2, OF0.4, OF0.6, RF0.2, RF0.4, and RF0.6. In this study, vibration data from two positions, A1 and A2, under 25.6 kHz frequency and 0N load are selected to construct the dataset. The acceleration signals for each condition are collected through the X, Y, and Z channels.

4.4.2 Comparative experiment

As Fig. 17 shows, compared to other models, the proposed model and Model DCA-BIGRU achieve higher accuracy across all five noise environments. Specifically, when SNR = −8, Model DCA-BIGRU achieves higher accuracy than the proposed model. However, when SNR = −10 and SNR = −2, the proposed model outperforms Model DCA-BIGRU in accuracy. In the remaining two noise environments, the accuracy difference between the proposed model and Model DCA-BIGRU is minimal. This demonstrates that the pro-

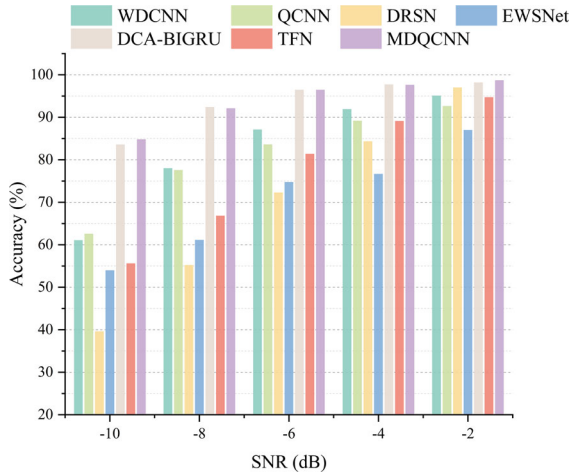


Fig. 17 Comparison of diagnostic accuracy of models in different noise environments

posed model still achieves high fault diagnosis accuracy on the Shandong University of Science and Technology bearing dataset, indicating its strong generalization capability.

5 Conclusion

This paper proposes a multi-scale quadratic convolutional neural network (MDQCNN) based on multi-sensor data fusion for bearing fault diagnosis. Firstly, vibration signals collected from sensors at different locations are used as inputs and fed into a time frequency filter for preprocessing. Among them, the time domain filter is constructed using a quadratic convolutional neural network, while the frequency domain filter is composed of a fully connected neural network. The filtered signal is input to a quadratic convolutional neural network, which uses convolution kernels of different sizes to extract low-frequency, mid frequency, and high-frequency features. Next, more effective features are further extracted through cross attention mechanism, and finally input into the classifier to complete fault diagnosis. The experimental results show that MDQCNN outperforms other compared models under different loads and noise disturbances, exhibiting strong generalization ability and robustness.

Future research can further explore the problem of fault diagnosis under various working conditions in practical engineering. In addition, the multi-sensor fusion method proposed in this paper is mainly based on vibration signals, which originate from the analysis of bearing failure vibration mechanisms and may not be directly applicable to other types of mechanical equipment or sensor systems. These issues will become the focus of future research.

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Data availability The data used in this study is the international universal data set.

Declarations

Conflict of interest The authors declare that they have no known conflict of interest or personal relationships that could have appeared to influence the work reported in this paper.

References

1. Li, S., Peng, Y., Bin, G.: Prediction of wind turbine blades icing based on CJBM with imbalanced data. IEEE Sens. J. **23**, 19726–19736 (2023)
2. Jiang, X., Li, X., Wang, Q., et al.: Multi-sensor data fusion-enabled semi-supervised optimal temperature-guided PCL framework for machinery fault diagnosis. Inf. Fusion **101**, 102005 (2024)
3. Shao, H., Li, W., Cai, B., et al.: Dual-threshold attention-guided GAN and limited infrared thermal images for rotating machinery fault diagnosis under speed fluctuation. IEEE Trans. Ind. Inf. **19**(9), 9933–9942 (2023)
4. Singh, V., Gangsar, P., Porwal, R., et al.: Artificial intelligence application in fault diagnostics of rotating industrial machines: a state-of-the-art review. J. Intell. Manuf. **34**(3), 931–960 (2023)
5. Yan, X., Yan, W.J., Xu, Y., et al.: Machinery multi-sensor fault diagnosis based on adaptive multivariate feature mode decomposition and multi-attention fusion residual convolutional neural network. Mech. Syst. Signal Process. **202**, 110664 (2023)
6. Xiao, Y., Shao, H., Feng, M., et al.: Towards trustworthy rotating machinery fault diagnosis via attention uncertainty in transformer. J. Manuf. Syst. **70**, 186–201 (2023)
7. Lin, J., Shao, H., Zhou, X., et al.: Generalized MAML for few-shot cross-domain fault diagnosis of bearing driven by heterogeneous signals. Expert Syst. Appl. **230**, 120696 (2023)
8. Wang, S., Lei, Y., Yang, B., et al.: A graph neural network-based data cleaning method to prevent intelligent fault diagnosis from data contamination. Eng. Appl. Artif. Intell. **126**, 107071 (2023)
9. Ye, M., Yan, X., Jiang, D., et al.: MIFDELN: a multi-sensor information fusion deep ensemble learning network for diagnosing bearing faults in noisy scenarios. Knowl. Based Syst. **284**, 111294 (2024)
10. Zhang, F., Yin, Z., Xu, F., et al.: MAB-DrNet: bearing fault diagnosis method based on an improved dilated convolutional neural network. Sensors **23**(12), 5532 (2023)
11. Fan, C., Peng, Y., Shen, Y., et al.: Variable scale multilayer perceptron for helicopter transmission system vibration data abnormity beyond efficient recovery. Eng. Appl. Artif. Intell. **133**, 108184 (2024)
12. Lei, Y., Yang, B., Jiang, X., et al.: Applications of machine learning to machine fault diagnosis: a review and roadmap. Mech. Syst. Signal Process. **138**, 106587 (2020)
13. Li, R., He, D.: Rotational machine health monitoring and fault detection using EMD-based acoustic emission feature quantification. IEEE Trans. Instrum. Meas. **61**(4), 990–1001 (2012)
14. Si, Y., Wang, Y., Zhou, D.: Key-performance-indicator-related process monitoring based on improved kernel partial least squares. IEEE Trans. Ind. Electron. **68**(3), 2626–2636 (2020)
15. Yan, X., Yan, W., Yuen, K.V., et al.: An adaptive variational mode extraction method based on multi-domain and multi-objective optimization for bearing fault diagnosis. Struct. Health Monit. **22**(4), 2708–2733 (2023)

16. Miao, Y., Li, C., Shi, H., et al.: Deep network-based maximum correlated kurtosis deconvolution: a novel deep deconvolution for bearing fault diagnosis. Mech. Syst. Signal Process. **189**, 110110 (2023)
17. Li, S., Peng, Y., Shen, Y., et al.: Rolling bearing fault diagnosis under data imbalance and variable speed based on adaptive clustering weighted oversampling. Reliab. Eng. Syst. Saf. **244**, 109938 (2024)
18. Yan, S., Shao, H., Min, Z., et al.: FGDAE: a new machinery anomaly detection method towards complex operating conditions. Reliab. Eng. Syst. Saf. **236**, 109319 (2023)
19. Liang, P., Li, Y., Wang, B., et al.: Remaining useful life prediction via a deep adaptive transformer framework enhanced by graph attention network. Int. J. Fatigue **174**, 107722 (2023)
20. Xun, Y., Jiang, Q., Li, S., et al.: Digital twin-driven discriminative graph learning networks for cross-domain bearing fault recognition. Comput. Ind. Eng. **193**, 110292 (2024)
21. Yan, X., Lu, Y., Liu, Y., et al.: Attention mechanism-guided residual convolution variational autoencoder for bearing fault diagnosis under noisy environments. Meas. Sci. Technol. **34**(12), 125046 (2023)
22. Zou, J., Lou, J., Wang, B., et al.: A novel deep reinforcement learning based automated stock trading system using cascaded lstm networks. Expert Syst. Appl. **242**, 122801 (2024)
23. Shao, H., Jiang, H., Zhang, H., et al.: Electric locomotive bearing fault diagnosis using a novel convolutional deep belief network. IEEE Trans. Ind. Electron. **65**(3), 2727–2736 (2017)
24. Zhao, X., Yao, J., Deng, W., et al.: Intelligent fault diagnosis of gearbox under variable working conditions with adaptive intraclass and interclass convolutional neural network. IEEE Trans. Neural Netw. Learn. Syst. **34**(9), 6339–6353 (2022)
25. Zhang, W., Peng, G., Li, C., et al.: A new deep learning model for fault diagnosis with good anti-noise and domain adaptation ability on raw vibration signals. Sensors **17**(2), 425 (2017)
26. Yu, W., Zhao, C.: Broad convolutional neural network based industrial process fault diagnosis with incremental learning capability. IEEE Trans. Ind. Electron. **67**(6), 5081–5091 (2019)
27. Qin, Z., Wang, Y., Yuan, Z., et al.: Triboelectric linear bearing sensor for self-powered condition monitoring using wavelet transform and lightweight CNN. Sens. Actuators A **359**, 114455 (2023)
28. He, C., He, D., Wei, Z., et al.: A train bearing imbalanced fault diagnosis method based on extended CCR and multi-scale feature fusion network. Nonlinear Dyn. **112**, 1–27 (2024)
29. Xu, Y., Shu, R., Li, S., et al.: Imbalanced learning for gearbox fault detection via attention-based multireceptive field convolutional neural networks with an adaptive label regulation loss. IEEE Trans. Instrum, Meas (2024)
30. Li, S., Peng, Y., Bin, G., et al.: Research on bearing fault diagnosis method based on cjbm with semi-supervised and imbalanced data. Nonlinear Dyn. **112**(22), 19759–19781 (2024)
31. Jiang, L., Li, X., Wu, L., et al.: Bearing fault diagnosis method based on a multi-head graph attention network. Meas. Sci. Technol. **33**(7), 075012 (2022)
32. Wei, Q., Tian, X., Cui, L., et al.: WSAFormer-DFFN: a model for rotating machinery fault diagnosis using 1D window-based multi-head self-attention and deep feature fusion network. Eng. Appl. Artif. Intell. **124**, 106633 (2023)
33. Li, S., Ji, J.C., Xu, Y., et al.: IFD-MDCN: multibranch denoising convolutional networks with improved flow direction strategy for intelligent fault diagnosis of rolling bearings under noisy conditions. Reliab. Eng. Syst. Saf. **237**, 109387 (2023)
34. Huang, Y.J., Liao, A.H., Hu, D.Y., et al.: Multi-scale convolutional network with channel attention mechanism for rolling bearing fault diagnosis. Measurement **203**, 111935 (2022)
35. Huo, T., Deng, L., Zhang, B., et al.: Novel imbalanced subdomain adaption multiscale convolutional network for cross-domain unsupervised fault diagnosis of rolling bearings. Meas. Sci. Technol. **35**(1), 015905 (2023)
36. Niazi, S.G., Huang, T., Zhou, H., et al.: Multi-scale time series analysis using TT-ConvLSTM technique for bearing remaining useful life prediction. Mech. Syst. Signal Process. **206**, 110888 (2024)
37. Guan, Y., Meng, Z., Sun, D., et al.: 2MNet: multi-sensor and multi-scale model toward accurate fault diagnosis of rolling bearing. Reliab. Eng. Syst. Saf. **216**, 108017 (2021)
38. Zhu, Y., Liang, X., Wang, T., et al.: Multi-information fusion fault diagnosis of bogie bearing under small samples via unsupervised representation alignment deep Q-learning. IEEE Trans. Instrum. Meas. **72**, 1–15 (2022)
39. Xu, Y., Ji, J.C., Ni, Q., et al.: A graph-guided collaborative convolutional neural network for fault diagnosis of electromechanical systems. Mech. Syst. Signal Process. **200**, 110609 (2023)
40. Fan, F., Cong, W., Wang, G.: A new type of neurons for machine learning. Int. J. Numer. Methods Biomed. Eng. **34**(2), e2920 (2018)
41. Fan, F., Wang, G.: Fuzzy logic interpretation of quadratic networks. Neurocomputing **374**, 10–21 (2020)
42. Liao, J.X., Dong, H.C., Sun, Z.Q., et al.: Attention-embedded quadratic network (qttention) for effective and interpretable bearing fault diagnosis. IEEE Trans. Instrum. Meas. **72**, 1–13 (2023)
43. Liao, J.X., He, C., Li, J., et al.: Classifier-guided neural blind deconvolution: a physics-informed denoising module for bearing fault diagnosis under noisy conditions. Mech. Syst. Signal Process. **222**, 111750 (2025)
44. Van der Maaten, L., Hinton, G.: Visualizing data using t-SNE. J. Mach. Learn. Res. **9**(11), 2579–2605 (2008)
45. Zhao, M., Zhong, S., Fu, X., et al.: Deep residual shrinkage networks for fault diagnosis. IEEE Trans. Ind. Inf. **16**(7), 4681–4690 (2019)

46. He, C., Shi, H., Si, J., et al.: Physics-informed interpretable wavelet weight initialization and balanced dynamic adaptive threshold for intelligent fault diagnosis of rolling bearings. J. Manuf. Syst. **70**, 579–592 (2023)
47. Zhang, X., He, C., Lu, Y., et al.: Fault diagnosis for small samples based on attention mechanism. Measurement **187**, 110242 (2022)
48. Chen, Q., Dong, X., Tu, G., et al.: TFN: an interpretable neural network with time-frequency transform embedded for intelligent fault diagnosis. Mech. Syst. Signal Process. **207**, 110952 (2024)
49. Fang, H., Deng, J., Bai, Y., et al.: CLFormer: a lightweight transformer based on convolutional embedding and linear self-attention with strong robustness for bearing fault diagnosis under limited sample conditions. IEEE Trans. Instrum. Meas. **71**, 1–8 (2021)
50. Wang, J., Zhang, X., Zhang, Z., et al.: Attention guided multi-wavelet adversarial network for cross domain fault diagnosis. Knowl. Based Syst. **284**, 111285 (2024)

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